

Measurement Techniques to Assess Accuracy of Volunteered Geographic Information

1. Introduction

Historically, geospatial data has been created and shared from a select group of organizations. In the United States, agencies like the USGS have been responsible for collecting geographic information and disseminating that information in both paper and digital formats. The relative paucity of data collectors was due to the high cost associated with collecting geographic data. This high cost included the technical training necessary to map objects in the field, the cost of the mapping equipment, and the time necessary to complete mapping projects. In recent years, the high cost associated with mapping has significantly decreased. Handheld GPS units are relatively inexpensive and they are now included on most cellular phones sold. Opensource and free versions of mapping software have been produced that provide much of the functionality of commercial geographic information systems. And the increase in availability of broadband internet access has allowed for data to be shared quickly and easily. By lowering the cost of entry for creating and sharing geographic information, mapping has become another addition of the Web 2.0 movement that allows not only the use of data shared over the internet, but also the creation of data. This melding of Web 2.0 and geography has resulted in what Goodchild refers to as “volunteered geographic information” (VGI) (2007). While this democratization of geospatial data has many potential benefits, including free access to data, objects mapped that were historically too insignificant to bother mapping, and the ability to quickly update maps as new data is collected, it is not without problems. Through much of history, quality of geospatial data was assumed to be relatively high. This was due to the nature of the provider, the US government in many cases, and the time and effort that went into collecting data. Now, however, with the ability for anyone, regardless of training or expertise, to create geographic data, that notion of assumed quality has disappeared. The person creating the data may be a retired navigation officer with military training or a 13 year old with an iPhone. As a result, researchers today are applying techniques for estimating accuracy to VGI in an effort to determine the quality of information that is produced by volunteers.

The term “accuracy” has undergone a series of revisions when used in respect to geographic data. Originally, geospatial data was often compared to its real counterpart on the earth’s surface, however, a consensus that this task was impossible for a variety of reasons led to a shift to relative accuracy instead (Goodchild and Hunter, 1997). Now techniques are used to compare the accuracy of one data set to another. In terms of geospatial data, accuracy is comprised of many components. Van Oort (2006) lists 8 different components: lineage, positional accuracy, attribute accuracy, logical consistency, completeness, semantic accuracy, usage, purpose and constraints, and temporal quality. Researchers interested in the accuracy of VGI have mostly focused on positional accuracy, attribute accuracy, and completeness. The remainder of this paper

will be used to present and discuss some of the techniques that are currently being used to evaluate the relative accuracy of geospatial data followed by specific cases involving volunteered geographic information.

2. Techniques for Estimating Geospatial Data Quality

The following articles describe various techniques used to estimate accuracy of geospatial data.

Goodchild and Hunter (1997) and Hunter (1999) describe a technique for estimating the positional accuracy of line features. Rather than compare the positional accuracy of the data to its real-world counterparts, the authors propose a relative comparison between a data set of unknown quality (test source) and one of high quality (reference source). The authors' method consists of creating a buffer around the reference source and determining what percentage of the test source falls within this buffer. The authors do not mention a specific technique to accomplish this analysis, but within the ArcGIS environment, the buffer tool could be used to create the buffer around the reference source while an intersect or union tool could be used to split the test source into regions within the buffer and outside the buffer. Finally, a spatial join or summarize function could be used to calculate the total length of the feature within and outside of the buffer. The user changes the buffer width in order to capture a higher or lower percentage of the test source. This technique is recommended when using natural features that do not have obvious nodes (streams or coastlines compared to road networks). This technique benefits from being easy to implement within a GIS and produces easy to interpret results. This technique is best suited when comparing a limited number of reference sources to test sources, for instance, when comparing interstate data. This technique may present problems when comparing two entire line networks because the positional accuracy of the test source may be very displaced. So even though two roads overlap, they may not be the same road.

Tveite (1999) recommends several techniques for assessing the positional accuracy between lines as well as the shape fidelity between lines. For estimating the positional point accuracy of a line, the author recommends determining the distance between corresponding points (the same street intersection for instance) within a reference and test data set using a variation of Euclidean distance. However, if there are no obvious nodes (a stream or coastline for instance), the author recommends starting at the first node of the reference data, moving down the line a set distance, extracting the coordinates at that distance, searching for the closest node on the test source, and comparing those two values. This process is repeated until the end of the reference source is reached. The author does not specifically describe the tools that are used, but a combination of network analyst and distance tools is probably appropriate. This technique may present problems because the two data sets may be near each other, but that does not necessarily mean that the node in the reference set represents the same node in the test source. The author describes a different technique to use when comparing the general shape of two data sets. Building on the Goodchild and Hunter (1997) technique, the author recommends building a buffer around both the test and reference source and

comparing what percentage of the buffers overlay. The author states that this method is best suited for data sets of homogenous quality; however, this technique will be shown below to be used in data sets of heterogeneous quality.

Van Niel and McVicar (2002) assessed the positional accuracy of an authoritative data set in order to determine if it met stated positional accuracy requirements and to determine if any bias was present in the positional error of the data. The authors combined previous work in assessing positional accuracy of linear networks. The authors used the point positional technique of Tveite (1999) using intrinsic nodes in the data sets. The authors refer to this technique as absolute positional error because the user is comparing corresponding nodes in the two data sets. The authors also used the Goodchild and Hunter (1997) technique of line buffering to assess what they refer to as functional positional error. This is an important distinction between the point positional error technique because this technique simply measures the distance between the test data source and the reference data source, not if the distance is between corresponding nodes. The authors found that absolute positional error was two times greater than functional positional error and not interchangeable.

Lee (2009) compared the positional accuracy of US Census 2000 TIGER line files to commercially available data as well as USGS data. The author compared TIGER street network positional accuracy to commercially available Geography Data Technology Dynamap data by comparing the locations of points between the data sets. However, rather than compare intrinsic nodes, like intersections, the author compared geocoded points created by the two data sets. The author determined the distance between the two sets of points using a formula for calculating the great circle distance based on spherical trigonometry. To take into account problems with geocoding itself, the author only used points that resulted in a match score of 100 or were unique matches. The author also compared TIGER stream line files to those of the US GNIS. The author converted the latitude and longitude of the GNIS data into shapefiles and used the near tool to determine the distance from the GNIS data to the stream data. I was under the impression that GNIS data was simply points. If that is the case, the author does not state whether the GNIS data represents the head, mouth, or interior point of the stream. Like the technique presented by Tveite (1999), this technique may not be comparing corresponding nodes between the data sets and may instead represent functional accuracy as defined by Van Niel and McVicar (2002).

3. Techniques Employed for Estimating Geospatial Data Quality in VGI

The following articles illustrate the application of the techniques listed above in scenarios specific to volunteered geographic information.

The first researcher to publish quantitative analysis of the positional accuracy of volunteered geographic information was Haklay from the University College London. It is important to note that this specific application of GIS is a relatively new, niche field and much of the published material is from Haklay or his students. His articles cite his own research and the research, often unpublished or without peer review in the form of

theses at the masters and PhD level, of his own students to a large degree. I do not feel that this reflects poorly on Haklay, but rather that this particular field of GIScience has not experienced widespread interest yet and requires further analysis from a broader range of researchers.

Haklay's (2008) first published article on VGI and quality attempts to measure both completeness and positional accuracy of OpenStreetMap (OSM) data compared to authoritative data from the Ordnance Survey in England. Haklay employed the Goodchild and Hunter (1997) buffer technique to compare the functional accuracy of motorways (the equivalent of interstates in the US), A-roads (highways), and B-roads (major roads) created by the Ordnance Survey and OSM. The author compared each dataset independently. In addition to this vector technique, the author also compared OSM data to raster data in the form of Ordnance Survey tiles (equivalent of USGS topographic maps). The author randomly selected 5 tiles within London and 100 sample locations per tile. The author does not specify if he is comparing motorways, A-roads, or B-roads, nor does he specify if the sample locations are randomly selected or stratified. This technique was used to compare the intersections (nodes) of the OSM data to an authoritative data set and establish absolute positional error. The author does not describe which technique was used to measure the distance between the two data sets. To determine completeness of the data sets, the author divides England into a set of 1 km^2 grids. Within each grid the author determines the length of road segments using a series of SQL queries. The author specifically mentions clipping each grid in MapInfo.

Haklay, Basiouka, Antoniou, and Ather (2010) also developed a technique to estimate the accuracy of the OSM data using intrinsic metadata. The authors wanted to determine if the number of contributors to a certain region could be correlated with data quality. The authors employed the same techniques described in Haklay (2008) to measure functional positional error of the data by breaking the study tiles, four randomly selected 25 km^2 tiles in London, into 1 km^2 grids, clipping the motorways, A-roads, and B-roads within each grid, and determining the average positional accuracy within each grid. Next, the authors compared the average positional accuracy within each grid to the number of contributors per grid. The authors also developed an algorithm to automatically find corresponding nodes between the two data sets to be used for all of England. The authors do not specify which technique is used to measure the distance between corresponding nodes.

Haklay and Ellul's (2010) article combines Haklay's previous work with an index of socio-economic status called the Index of Deprivation in an effort to determine if there is a socioeconomic bias in the completeness of OpenStreetMap data compared to Ordnance Survey data. The authors began by creating a grid of 1 km^2 cells for all of England. Then the road data from the Ordnance Survey and OSM is split along the boundaries of the grid. Then a sum for the road lengths is calculated per grid. The authors use an algorithm with a spatial query to complete these steps, probably something like a spatial join with the sum box checked in Arc. Finally, the authors divide the length of the OSM data by the Ordnance Survey data to calculate a percentage which represents completeness. Using the same 1 km^2 grids, the authors also calculate the percentage of

OSM data that has attribute data (name and road type) and compare that to the Ordinance Survey data as a percent.

Zielstra and Zipf (2010) used Haklay's (2008) methods to compare completeness of OpenStreetMap data to Tele Atlas data for Germany. It is assumed that if the total length of OSM data is greater than the total length of Tele Atlas data, then OSM is more complete (because OSM also contains walking paths, bicycle trails, and other features that are not typically mapped by commercial entities like Tele Atlas). If Tele Atlas data is longer than OSM data then Tele Atlas is assumed to be more complete. The authors began by comparing the total length of the two data sets for all of Germany. They next compared the data sets within the 5 largest cities in Germany and 5 medium sized cities using some type of administrative boundary to clip their data to. Next, the authors created 1 km² grids for all of Germany and used Haklay and Ellul's (2010) technique for measuring completeness per grid cell. Lastly, the authors compared the length of road networks within varying buffers centered on each city center from the above analyses. For large cities the authors used buffers of 0-10 km, 10-25 km, 25-50 km and for medium sized cities the authors used buffers of 0-5 km, 5-15 km, and 15-25 km. The authors used the same technique as above, but instead of clipping the road boundaries to the grid cells, they used the boundaries of the various buffer sizes.

Girres and Touya (2010) attempted a much more comprehensive analysis of the different components of quality than any of the other authors represented in this paper. The authors attempted to measure eight different components: positional accuracy, attribute accuracy, semantic accuracy, completeness, logical consistency, temporal accuracy, usage, and lineage. The authors used a relative measure of accuracy for much of the analysis between French OpenStreetMap and commercial data available from BD Topo. For point positional accuracy, the authors calculated the Euclidean distance between homologous pairs of road intersection points. For positional accuracy of linear data, the authors compared homologous lengths of road and coastline using the Hausdorff distance and average distance formulas. For positional accuracy of polygonal data, the authors used a measure of surface distance, which appears to be a measure of how much two polygons overlap. The authors also compared granularity, what is the shortest distance between nodes in a polygon, and compactness using Miller's measure. Attribute accuracy was calculated in two ways. The authors first determined the percentage of objects in OSM with tags and then used a Levenstein distance of strings to determine how similar the tags were between corresponding objects. Semantic accuracy was calculated by comparing the tags of OSM data to the corresponding objects' tags in BD Topo data and calculating a percent. Completeness was calculated by comparing the total length/area of objects in the OSM data set to the BD Topo data as well as performing a count of the total objects in both data sets. Logical consistency was measured by determining the connectivity of the OSM road network. Temporal accuracy was measured by graphing the number of objects that had been added to the OSM data set over time. Lineage was calculated by determining the percentage of objects with source tags. This article seemed to attempt to accomplish too much at one time and as a result some of the analysis techniques seem weak in comparison to other researchers. For instance, the authors claim to measure temporal accuracy by plotting the number of

features that are added to the data set over time. This simply tells how many features have been added, it does not indicate how long the feature was present before it was added to the OSM dataset. Also, the authors fail to give any detailed methods of how they used GIS to actually perform their analysis other than sharing a formula from an equation. This article may benefit from being separated into multiple articles where each component of accuracy can receive more attention and detail.

4. Discussion

The articles present in this paper represent both broad techniques used to measure different components of geospatial data accuracy as well as specific case studies involving volunteered geographic information. The authors use a variety of techniques as well as variations of the same technique to measure different components of accuracy. The majority of the authors measured positional accuracy of point and line data and completeness of features and attributes. For positional accuracy of point features, many of the authors used a variation of Euclidean distance. For positional accuracy of linear features, the authors used a combination of the buffer tool, a union or intersect, and then a technique to summarize the length of features within and outside of the buffer. To measure completeness of linear features, most authors employed a summary of feature length within various polygons. The polygons consisted of administrative boundaries, buffers of points, and grids. To accomplish this task the authors used either a clip tool on the line features for each polygon of interest or a spatial join while simultaneously summing the line length per polygon. To estimate completeness of attributes the authors used SQL queries like select by attribute. Two authors appeared to use network topology to measure connectivity or distance along a network which may have included Network Analyst. Some authors used specific formulas in their calculations without giving much detail in how the calculation was performed. If the calculation was performed in the ArcGIS environment, then it was probably conducted using the field calculator or exported to Excel. Most authors used a combination of tools to accomplish their desired goal because one tool will typically not perform all needed tasks. ModelBuilder within ArcGIS may have been used to efficiently join different tools together while some authors specifically mentioned algorithms without describing the environments those algorithms were built or run in.

Based on my current level of GIS knowledge, I was prepared to understand the majority of tools and techniques used within these articles. While some authors did not specifically outline their detailed methodology or used vague terminology, I would still be able to perform the same task. Many of the authors used various forms of statistics to determine if their results were significant without mentioning which tests were performed. While we used some statistical tests throughout the course, I would not be able to duplicate the authors' statistical works without further training. Other authors used a variety of formulas to calculate certain values pertaining to distance and area. These formulas were limited in use but it still may be beneficial to learn about their appropriate use and execution on spatial data. While researchers have been studying accuracy of geospatial data for some time, the application of this field to VGI is still new. As a result, I cannot say that the techniques that are currently being used will prove comprehensive as more researchers add to this field.

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